



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO2_AT2 — Visualization Development Exercise

Name: A Roja

Reg No: 192424226

DSA0601-Data Handling and Visualization

Question 1 — Chart Selection (Topic 1: Directory of Visualizations)

Task: For each scenario below, name the single best chart type from the directory (Amounts / Distribution / Proportion / x–y Relationship / Geospatial / Network / Temporal / Multivariate) and justify your choice in 1–2 sentences. No code required for this question.

1. Comparing the population of 8 countries in the current year.
2. Showing how delivery times vary across 5 courier companies.
3. Showing the proportion of a household budget spent on rent, food, transport, and savings.
4. Showing the relationship between hours studied and exam marks for 50 students.
5. Showing average temperature for each month across 3 different cities at once.
6. Showing which airports are directly connected by flight routes.

Chart Selection

1. Comparing the population of 8 countries in the current year.

Best Chart Type: Amounts

An Amounts chart is suitable for comparing the population values of different countries. A bar chart clearly shows the differences in population and identifies the country with the highest or lowest population.

2. Showing how delivery times vary across 5 courier companies.

Best Chart Type: Distribution

A Distribution chart is appropriate for showing how delivery times are spread and how much they vary across the five courier companies. A box plot can compare the range, median, and variation of delivery times.

3. Showing the proportion of a household budget spent on rent, food, transport, and savings.

Best Chart Type: Proportion

A Proportion chart is suitable because each category represents a part of the total household budget. It clearly shows the percentage or share spent on rent, food, transport, and savings.

4. Showing the relationship between hours studied and exam marks for 50 students.

Best Chart Type: x-y Relationship

An x-y Relationship chart is best for examining the relationship between hours studied and exam marks. A scatter plot can show whether an increase in study hours is associated with higher exam marks.

5. Showing average temperature for each month across 3 different cities at once.

Best Chart Type: Multivariate

A Multivariate chart is suitable because the visualization involves temperature, month, and three different cities. It allows the temperature patterns of all three cities to be compared simultaneously.

6. Showing which airports are directly connected by flight routes.

Best Chart Type: Network

A Network chart is the best choice because airports can be represented as nodes and flight routes as connections between them. It clearly shows which airports are directly connected to one another.

Question 2 — Bar Plot (Topic 2: Amounts)

Dataset — app_downloads.csv

App_Category	Downloads_Millions
Social Media	58
Gaming	46
Productivity	27
Finance	19
Health & Fitness	33
Education	24

Task:

- Create a horizontal bar plot of Downloads_Millions by App_Category, sorted in ascending order.
- Use a color gradient (single hue, varying shade) so bar color intensity reflects download count.
- Add the exact value as a text label at the end of each bar.

Python: use pandas to build the DataFrame and matplotlib.pyplot.barh (or seaborn.barplot with orient="h").

R: use a data.frame, ggplot2::geom_col() with coord_flip() (or geom_col() plus aes(y = reorder(...))).

Explanation:

1. Social Media has the highest number of downloads, with 58 million downloads.
2. Gaming has the second-highest downloads, with 46 million.
3. Health & Fitness has 33 million downloads.
4. Finance has the lowest number of downloads, with 19 million.
5. The horizontal bar chart makes it easy to compare downloads across all app categories.

R Implementation

```
library(ggplot2)
```

```

df <- data.frame(
  App_Category = c(
    "Social Media",
    "Gaming",
    "Productivity",
    "Finance",
    "Health & Fitness",
    "Education"
  ),
  Downloads_Millions = c(58, 46, 27, 19, 33, 24)
)
df$App_Category <- factor(
  df$App_Category,
  levels = df$App_Category[order(df$Downloads_Millions)]
)
ggplot(df, aes(
  x = App_Category,
  y = Downloads_Millions,
  fill = Downloads_Millions
)) +
  geom_col() +
  geom_text(
    aes(label = Downloads_Millions),
    hjust = -0.2
  ) +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  coord_flip() +
  labs(
    title = "App Downloads by Category",

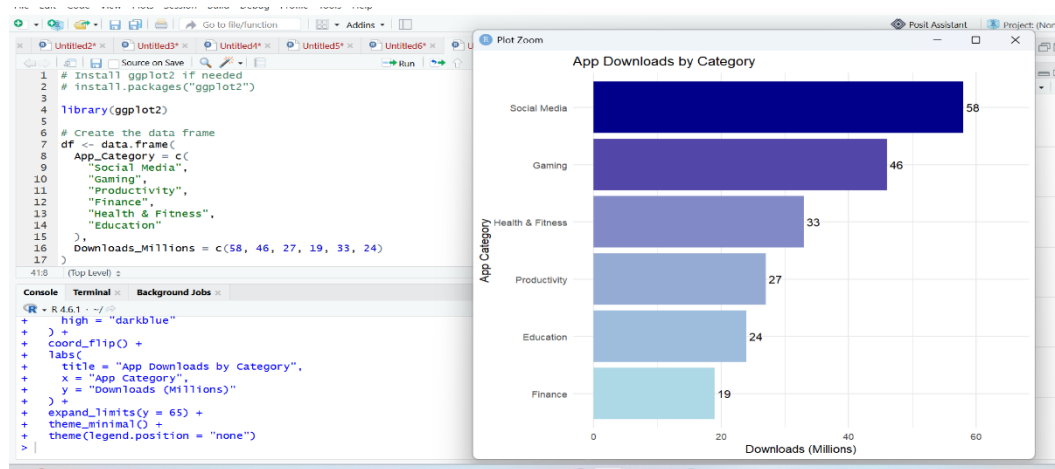
```

```

x = "App Category",
y = "Downloads (Millions)"
) +
expand_limits(y = 65) +
theme_minimal() +
theme(legend.position = "none")

```

OUTPUT:



Question 3 — Grouped/Stacked Bars, Dot Plot & Heatmap (Topic 2: Amounts)

Dataset — website_traffic.csv

Device	Month	Visits_000s
Desktop	Jan	45
Desktop	Feb	48
Desktop	Mar	42

Desktop	Apr	50
Mobile	Jan	80

Mobile	Feb	88
Mobile	Mar	95
Mobile	Apr	102
Tablet	Jan	20
Tablet	Feb	18
Tablet	Mar	17
Tablet	Apr	15

Task (produce all four views from the same dataset):

7. Grouped bar chart: Visits_000s by Month, grouped by Device.

8. Stacked bar chart: same data, stacked by Device within each Month, so each bar shows total monthly traffic.

9. Dot plot: total 4-month traffic per device (sum across months), ranked from

highest to lowest. 10. Heatmap: Device (rows) × Month (columns), color-encoded by Visits_000s.

Python: pandas.pivot_table to reshape; matplotlib/seaborn for grouped/stacked bars, seaborn.heatmap for the heatmap, matplotlib.pyplot.scatter (or plt.plot with markers) for the dot plot.

R: tidyr::pivot_wider where needed; ggplot2::geom_col(position = "dodge") for grouped, position = "stack" for stacked, geom_point() for the dot plot, geom_tile() for the heatmap.

R Implementation:

```
# Install packages if needed
# install.packages("ggplot2")
# install.packages("tidyr")
```

```
library(ggplot2)
library(tidyr)
df <- data.frame(
  Device = c(
```

```

"Desktop", "Desktop", "Desktop", "Desktop",
"Mobile", "Mobile", "Mobile", "Mobile",
"Tablet", "Tablet", "Tablet", "Tablet"
),
Month = c(
  "Jan", "Feb", "Mar", "Apr",
  "Jan", "Feb", "Mar", "Apr",
  "Jan", "Feb", "Mar", "Apr"
),
Visits_000s = c(
  45, 48, 42, 50,
  80, 88, 95, 102,
  20, 18, 17, 15
)
)
df$Month <- factor(
  df$Month,
  levels = c("Jan", "Feb", "Mar", "Apr")
)
# 1. GROUPED BAR CHART
ggplot(df, aes(x = Month, y = Visits_000s, fill = Device)) +
  geom_col(position = "dodge") +
  labs(
    title = "Monthly Website Visits by Device",
    x = "Month",
    y = "Visits (000s)"
  ) +
  theme_minimal()
# 2. STACKED BAR CHART
ggplot(df, aes(x = Month, y = Visits_000s, fill = Device)) +
  geom_col(position = "stack") +
  labs(
    title = "Stacked Monthly Website Visits by Device",
    x = "Month",
    y = "Visits (000s)"
  ) +
  theme_minimal()

# 3. DOT PLOT
total_traffic <- aggregate(
  Visits_000s ~ Device,
  data = df,
  sum
)

```

```

total_traffic <- total_traffic[
  order(-total_traffic$Visits_000s),
]
ggplot(
  total_traffic,
  aes(
    x = Visits_000s,
    y = reorder(Device, Visits_000s)
  )
) +
  geom_point(size = 4) +
  geom_text(
    aes(label = Visits_000s),
    hjust = -0.3
  ) +
  labs(
    title = "Total 4-Month Website Traffic by Device",
    x = "Total Visits (000s)",
    y = "Device"
  ) +
  xlim(0, 400) +
  theme_minimal()
# 4. HEATMAP
ggplot(
  df,
  aes(x = Month, y = Device, fill = Visits_000s)
) +
  geom_tile() +
  geom_text(
    aes(label = Visits_000s),
    color = "black"
  ) +
  labs(
    title = "Website Traffic Heatmap",
    x = "Month",
    y = "Device"
  ) +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  theme_minimal()

```

OUTPUT



Question 4 — Violin & Ridgeline Plots (Topic 3: Distributions)

Dataset — generate synthetically for reproducibility:

Route A: 50 delivery times (minutes) ~ Normal(mean=32, sd=6)

Route B: 50 delivery times (minutes) ~ Normal(mean=28, sd=4)

Route C: 50 delivery times (minutes) ~ Normal(mean=40, sd=9)

Route D: 50 delivery times (minutes) ~ Normal(mean=35, sd=5)

(Set a random seed of 7 in both Python and R so results are consistent across languages.) **Task:**

11. Violin plot: delivery-time distribution by route (all 4 routes on one plot), with an embedded boxplot or median marker.

12. Ridgeline plot: same 4 routes, overlapping density curves, ordered by mean delivery time (fastest at the bottom).

Python: `numpy.random.normal` to generate data, `seaborn.violinplot` for the violin plot, `joyplot.joyplot` (or a stacked `seaborn/matplotlib kdeplot FacetGrid`) for the ridgeline.

R: `rnorm()` to generate data, `ggplot2::geom_violin()` for the violin plot, `ggribes::geom_density_ridges()` for the ridgeline.

Violin plot

R Implementation

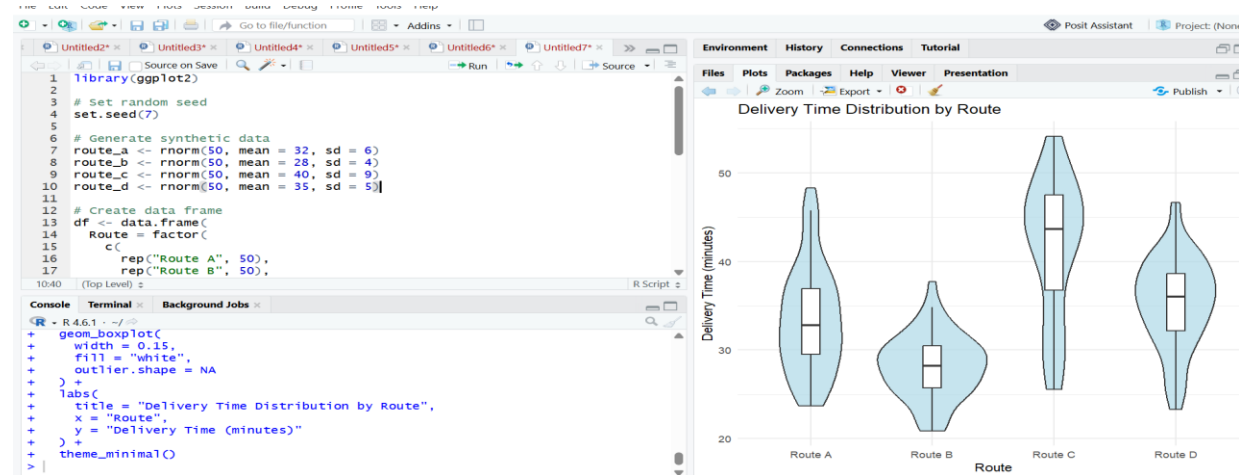
```
install.packages("ggplot2")
library(ggplot2)
set.seed(7)
route_a <- rnorm(50, mean = 32, sd = 6)
route_b <- rnorm(50, mean = 28, sd = 4)
route_c <- rnorm(50, mean = 40, sd = 9)
route_d <- rnorm(50, mean = 35, sd = 5)
df <- data.frame(
  Route = factor(
    c(
      rep("Route A", 50),
      rep("Route B", 50),
      rep("Route C", 50),
      rep("Route D", 50)
    ),
    levels = c("Route A", "Route B", "Route C", "Route D")
  ),
  Delivery_Time = c(
    route_a,
    route_b,
    route_c,
    route_d
  )
)
ggplot(df, aes(x = Route, y = Delivery_Time)) +
  geom_violin(fill = "lightblue", alpha = 0.7) +
  geom_boxplot(
    width = 0.15,
    fill = "white",
    outlier.shape = NA
  ) +
  labs(
```

```

title = "Delivery Time Distribution by Route",
x = "Route",
y = "Delivery Time (minutes)"
) +
theme_minimal()

```

OUTPUT



Ridgeline plot

```

install.packages("ggridges")
library(ggplot2)
library(ggridges)
set.seed(7)
route_a <- rnorm(50, mean = 32, sd = 6)
route_b <- rnorm(50, mean = 28, sd = 4)
route_c <- rnorm(50, mean = 40, sd = 9)
route_d <- rnorm(50, mean = 35, sd = 5)
df <- data.frame(
  Route = c(
    rep("Route A", 50),
    rep("Route B", 50),
    rep("Route C", 50),
    rep("Route D", 50)
  ),
  Delivery_Time = c(
    route_a,
    route_b,
    route_c,
    route_d
  )
)
df$Route <- factor(
  df$Route,

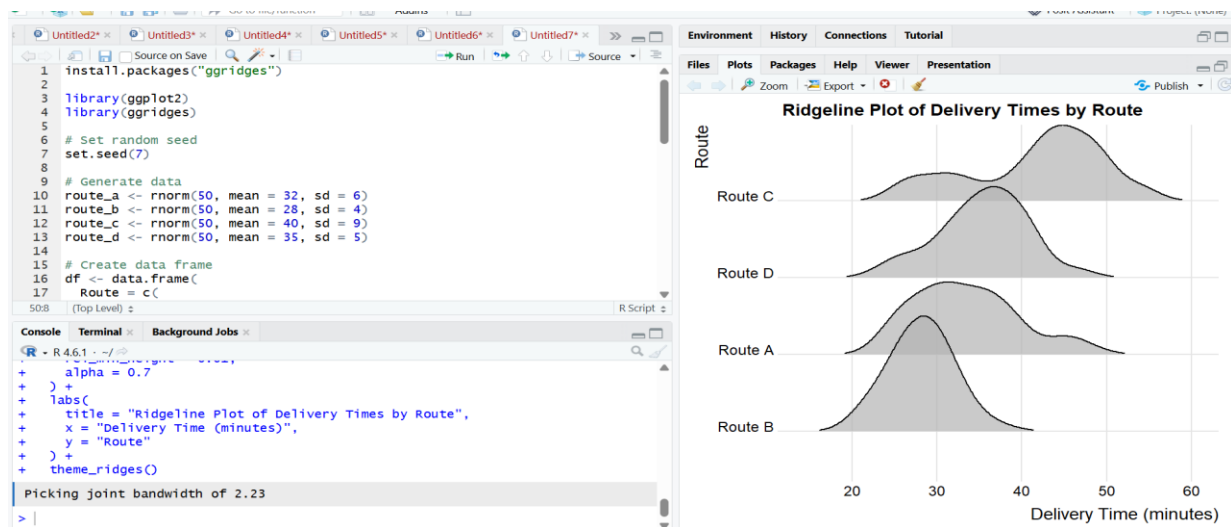
```

```

levels = c("Route B", "Route A", "Route D", "Route C")
)
ggplot(
  df,
  aes(
    x = Delivery_Time,
    y = Route
  )
) +
  geom_density_ridges(
    scale = 1.5,
    rel_min_height = 0.01,
    alpha = 0.7
  ) +
  labs(
    title = "Ridgeline Plot of Delivery Times by Route",
    x = "Delivery Time (minutes)",
    y = "Route"
  ) +
  theme_ridges()

```

OUTPUT



Question 5 — Histograms & Density Plots (Topic 3: Distributions)

Dataset: Use a built-in dataset available in both languages.

- Python: `seaborn.load_dataset("iris")` → use the `petal_length` column.
- R: the built-in `iris` dataset → use the `Petal.Length` column.

Task:

13. Plot a histogram of the chosen continuous variable with 20 bins.
14. Overlay a kernel density estimate (KDE) curve on the same plot.
15. Repeat both, split/colored by species (setosa, versicolor, virginica), to compare distribution shape across groups.

Python: `seaborn.histplot(..., kde=True)` or `matplotlib.pyplot.hist + scipy.stats.gaussian_kde`.

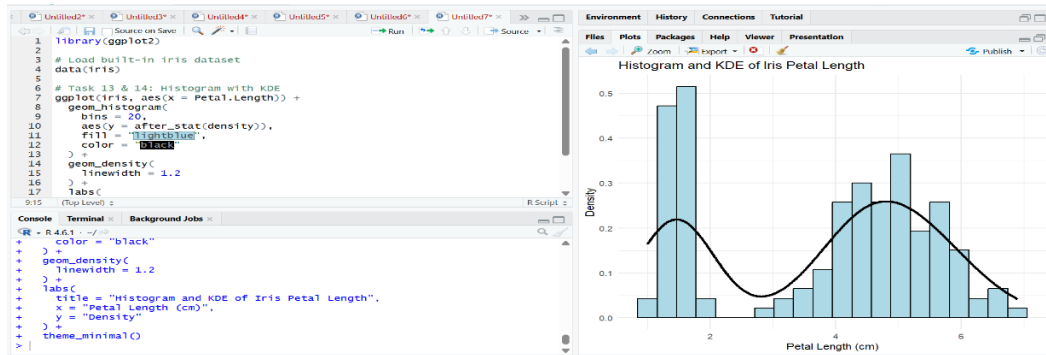
R: `ggplot2::geom_histogram()` combined with `geom_density()` (use `after_stat(density)` so scales match), with `fill =` mapped to Species for the grouped version.

Histogram with KDE Curve

```
# Install package if required
install.packages("ggplot2")
library(ggplot2)

# Use built-in iris dataset
# Histogram with KDE curve
ggplot(iris, aes(x = Petal.Length)) +
  geom_histogram(
    bins = 20,
    aes(y = after_stat(density)),
    fill = "lightblue",
    color = "black"
  ) +
  geom_density(
    linewidth = 1.2
  ) +
  labs(
    title = "Histogram and KDE of Iris Petal Length",
    x = "Petal Length (cm)",
    y = "Density"
  ) +
  theme_minimal()
```

OUTPUT:



Histogram and KDE by Species

```
library(ggplot2)
```

```
ggplot(
```

```
  iris,
```

```
  aes(
```

```
    x = Petal.Length,
```

```
    fill = Species
```

```
  )
```

```
)+
```

```
  geom_histogram(
```

```
    bins = 20,
```

```
    aes(y = after_stat(density)),
```

```
    alpha = 0.4,
```

```
    position = "identity"
```

```
  ) +
```

```
  geom_density(
```

```
    aes(color = Species),
```

```
    linewidth = 1.2
```

```
  ) +
```

```
  labs(
```

```
    title = "Petal Length Distribution by Iris Species",
```

```
    x = "Petal.Length (cm)",
```

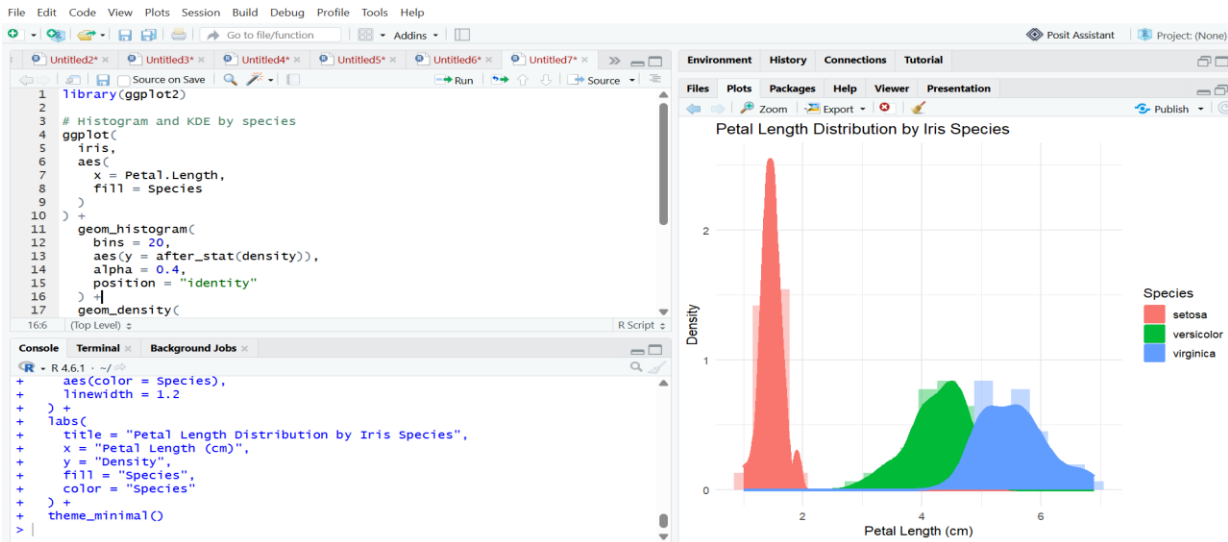
```
    y = "Density",
```

```

fill = "Species",
color = "Species"
) +
theme_minimal()

```

OUTPUT



Submission Checklist

- Q1: written chart-type justifications (no code)
- Q2: bar plot — Python + R
- Q3: grouped bar, stacked bar, dot plot, heatmap — Python + R
- Q4: violin plot, ridgeline plot — Python + R
- Q5: histogram + density (overall and grouped) — Python + R
- Each chart has a title, axis labels, and a short written interpretation